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November 16, 1992
SUPERSEDING
GGG-F-451C
June 13, 1968

FEDERAL SPECIFICATION

FLARING TOOL, TUBE HAND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand operated tools used for the purpose of flaring tubing.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Flaring tools shall be of the following types, classes, and styles, as specified (see 6.2).

Type I. Flaring, screw-actuated
- Class 1. Roller cone
  - Style A. Attached flare gage
  - Style B. Without attached gage
- Class 2. Swivel cone
  - Style A. Hinged dies
- Class 3. Combination, self-contained

Type II. Flaring, impact-actuated
- Class 1. Tube coupling
- Class 4. Hinged blocks

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40	-	Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standards:

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-376	-	Preferred Metric Units for General Use by the Federal Government

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 5120

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Military Specifications:

MIL-T-6845 - Tubing; Steel, Corrosion-Resisting, (18-8) Aircraft, Hydraulic System

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Military Standard Drawings:

MS 20819 - Sleeve, Flared Tube Fitting
MS 33583 - Tubing End - Standard Dimensions for Double Flare
MS 33584 - Tubing End - Standard Dimensions for Flared

Air Force-Navy Aeronautical Standard Drawings:

AN 815 - Union Flared Tube
AN 818 - Nut, Coupling
AN 929 - Cap Assembly - Pressure Seal Flared Tube Fitting

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

E 18 - Metallic Materials Rockwell Hardness and Rockwell Superficial Hardness of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Society of Automotive Engineers Standard (SAE):

SAE Handbook (Tube Fittings - Flared Tube)

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

American National Standards Institute (ANSI):

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of flaring tools otherwise conforming to this specification.

3.2 Materials. The materials used shall be such as to produce flaring tools conforming to the requirements specified hereinafter. Material shall be free from all defects and imperfections that may adversely affect the serviceability of the finished product.

3.3 Finish and coatings. All parts shall be neatly finished and free from pits, nodules, burrs, cracks, or other detrimental defects. Surfaces subject to corrosion or rust shall have any one or a combination of the following coatings unless a choice is specified (see 6.2):

(a) Chromium plate

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(b) Nickel plate
(c) Zinc plate
(d) Hot phosphating
(e) Oxide coating

3.4 Design. Tubing tools shall be of the latest design and shall conform to this specification. All tools and components shall withstand the stresses applied during the maximum normal operations without sign of failure or permanent deformation. Components of one tool must be interchangeable with those of any other tool of the same type, class, and style made by the same manufacturer.

3.5 Identification marking. All flaring tools shall be marked with the manufacturer's name or identifying symbol, the state or country of manufacture, and the size(s) of the tubing each tool will flare, unless otherwise specified. All marking shall be engraved, etched, molded, or stamped directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

3.6 Threads. All threads shall conform to the applicable requirements of FED-STD-H28.

3.7 Performance, hardness, and roughness tests. Tubing tools shall pass the performance concentricity, hardness, and surface roughness tests specified in 4.4.1, 4.4.2, and 4.4.3. When specified for Air Force purchases, they shall pass the additional tests specified in 4.4.4 for the tools indicated therein.

3.8 Repair parts. The successful bidders supplying flaring tools shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, and may be replaced or adjusted without requiring modification.

3.9 Carrying case. Unless otherwise specified, carrying cases or boxes shall be metal or blow-molded plastic. The carrying case shall be of sturdy construction, interior so configured to accommodate the flaring tool components and shall be designed to prevent excessive movement of the components when transported. Both inside and outside surfaces of metal cases or boxes shall have a protective coat of paint.

3.10 Operating instructions. Each tube shall have a set of operating instructions enclosed within the carrying case.

3.11 Type I, class 1, style A, flaring, screw-actuated, roller cone, attached flare gage. Style A flaring tools shall be suitable for producing high-quality flares having a 74° included angle (see 4.4.1.1). The tool shall flare cold-drawn corrosion-resisting steel tubing of the type specified in MIL-T-6845 having a yield point of 105,000 pounds per square inch, in sizes of 3/16, 1/4, 5/16, 3/8, 1/2, and 5/8 inch outside diameters. The attached flaring gage shall be easily operated, shall provide a positive means for flaring the tubing to the proper diameter, and shall be so secured that it will not become detached accidentally during normal use.

3.11.1 Type I, class 1 design. The tool shall be designed primarily for maintenance use, and flares shall be made with a minimum application of torque capable of being exerted by an average mechanic working in a restricted position such as inside an aircraft wing. The tool shall be capable of withstanding the harder usage common in field service and shall operate adequately without distortion or misalignment. The tool shall consist essentially of a die-block assembly, a compression assembly, and a metal carrying case. When specified (see 6.2), a yoke-mounted precision-feed control reamer for removing burrs from the inner edge of tubing prior to flaring shall be furnished.

3.11.2 Die block assembly. The assembly shall consist of two parallel bars, or a U-shaped bar, the necessary number of die blocks, and a tube clamping device. The die blocks shall maintain alignment when mounted between the parallel bars. Substantial rivet-type pins or studs shall hold the bars to the clamping device at one end and to a die block at the other end, when applicable. The clamping device shall have a swivel action to provide for removal of the tubing after flaring. The clamping screw shall have a hexagon extended washer head for tightening with a standard size wrench and the washer-head periphery shall be knurled for quick hand adjustment. The amount of potential squeeze between die blocks shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation. The die blocks shall have a hardness of 47 to 57 on the Rockwell C scale or shall be casehardened to a minimum depth of 0.003 inch and shall show a surface hardness of 84 to 91 on the Rockwell 15-N scale, or shall be casehardened to a minimum depth of 0.003 inch and shall show a Knoop Hardness Number (500 gm) of 575-775. Micro hardness testing is acceptable and shall be equivalent to the Rockwell Hardness requirements. The point of the clamping screw shall be rounded and hardened or shall have a hardened steel ball inserted to resist scoring under hard usage. The true radius of the tube-gripping holes shall have a tolerance of ± 0.0040 inch, and the surface roughness of the holes shall not exceed 100 microinch finish in order to minimize marring of the tubing during the flaring operation. The holes need not be countersunk as the flare is formed above the die blocks. The thickness of the blocks shall not exceed 1.020 inches so that flares may be made as close as possible to a bend in the tubing.

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3.11.3 Compression assembly. The compression assembly shall provide the means of applying the pressure necessary for flaring, and shall consist of a yoke, pressure sleeve and shaft assembly, and a roller flaring head.

3.11.3.1 Type I, class 1 yoke. The yoke shall have a threaded hole to receive the pressure sleeve, and construction shall permit ready alignment over the tube-gripping holes in the die blocks. Means shall be provided for mounting a device to gage the setting of the tube in the die blocks and controlling the amount of flare.

3.11.3.1.1 Yoke extension. A yoke extension shall be provided so that the yoke can be securely clamped in an ordinary bench vise.

3.11.3.2 Pressure sleeve and shaft assembly. The pressure sleeve shall have a thread conforming to FED-STD-H28, no coarser than the extra-fine thread series to fit the yoke and shall have a suitably reamed hole to provide a standard running fit for the shaft. The shaft shall have a hexagon extended washer head for turning with a standard-size wrench and the periphery of the washer shall be knurled for quick hand adjustment. A driving lug or pin shall be securely fastened to the underside of the shaft head so as to drive against a slot or a lug integral with, or securely fastened to, the pressure sleeve. After the downward feed of the pressure sleeve has flared the tubing to the specified diameter, it is necessary that the roller flaring head rotate in the same plane sufficiently to true-up the flare, and means shall be provided to assure the success of this operation. The roller flaring head shall be securely fastened to the small end of the shaft in such a manner as will preclude becoming detached during normal usage.

3.11.3.3 Roller flaring head. The flaring head shall have three or more rollers incorporated in such a manner that they rotate freely during the flaring operation. The rollers may be tapered or straight, but in either case, the surface shall have a maximum roughness of 16 microinches and the departure from straightness shall not exceed 0.002 inch per 1/2 inch length. The roller shall have a hardness of 60 to 65 on the Rockwell C scale. The rollers shall not become detached during normal usage. The flaring head shall have a surface hardness of 80 to 86 on the Rockwell 15-N scale.

3.11.4 Carrying case for type I, class 1, style A. Whether a deburring reamer is furnished or not, the case dimensions shall not exceed 6-1/4 inches maximum width, 13 inches maximum length by 2-1/2 inches maximum depth. The case shall have a hinged lid with either a positive or a firm friction catch.

3.11.5 Deburring reamer. If the deburring reamer is furnished (see 3.11.1 and 6.2), it shall be similar to the compression assembly specified under 3.11.3. The yoke shall be easily positioned over the tubing and shall be of the self-centering type. The sleeve shall have a thread conforming to FED-STD-H28, no coarser than the fine-thread series, shall have a reamed hole to pilot the reamer and have a knurled head for control of feed. The tapered reamer shall have sufficient flutes and shall be of adequate size to deburr the range of tubing specified for the flaring tool. The shaft shall be a running fit in the threaded sleeve, shall have a handle securely attached on one end, and shall be integral with or securely attached to the tapered reamer. The handle shall be of adequate design to enable efficient deburring of the range of tubing specified for the flaring tool. The reamer shall have a hardness of 55 to 62 on the Rockwell C scale.

3.11.6 Type I, class 1 weight. The weight of the complete tool with or without deburring reamer and carrying case shall not exceed 7 pounds.

3.11.7 Type I, class 1, style A, illustration. Type I, class 1, style A, flaring tools and deburring reamers shall be similar to Figure 1, (A) and (B), respectively.



(A) (B)

Figure 1. Type I, class 1, style A, flaring tool, screw-actuated, roller cone, attached flare gage.

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3.12 Type I, class 1, style B, flaring tool, screw-actuated, roller cone, without attached gage. Style B flaring tools shall be in accordance with the requirements for style A tools specified in 3.10 except for the following differences:

a. Tools shall flare fully annealed corrosion-resisting steel tubing, but are not required to flare it in the cold-drawn condition.
b. Capacity of tool shall be 3/4, 7/8, 1, and 1-1/4 inch outside diameter tubing.
c. Not furnished with a precision-feed control deburring reamer.
d. The thickness of the die blocks shall not exceed 1-1/2 inches.
e. Not furnished with an attached flaring gage.
f. Carrying case may exceed dimensions specified in 3.11.4 in order to adequately contain the tool.
g. Maximum limit of potential squeeze shall be 0.038 inch for 3/4 inch and larger tubing.

3.12.1 Type I, class 1, style B, illustration. Type I, class 1, style B, flaring tools shall be similar to Figure 2.

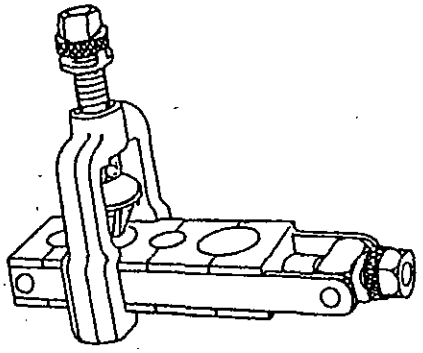


Figure 2. Type I, class 1, style B, flaring tool, screw-actuated, roller cone without attached gage.

3.13 Type I, class 2, style A, flaring tool, screw-actuated, swivel cone, hinged dies. Style A flaring tools shall produce 90° included-angle flares on soft steel, copper, and aluminum tubing to fit standard S.A.E. tube fittings. They shall consist of die bars and a compression assembly. When specified (see 6.2), they shall be furnished in boxes (see 3.9).

3.13.1 Double-lap flares. When specified (see 6.2), hardened adapter sets suitable for producing double-lap flares on 3/16, 1/4, 5/16, 3/8, 1/2, 5/8, and 3/4 inch tubing to fit standard S.A.E. tube fittings shall be furnished. The hardness shall comply with requirements specified in 3.12.3.3.

3.13.2 Type I, class 1, style B, die bars. The bars shall be hinged at one end to permit easy removal of the tubing after flaring. Wing nuts or other suitable means shall be provided to securely clamp the tubing during the flaring operation. The tube gripping holes shall be serrated or other means provided to prevent the tubing from slipping during the flaring operation and the holes shall be properly countersunk. The amount of potential squeeze between die blocks shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed the following limits: 0.040 inch for 5/8 inch and larger; 0.035 inch for 3/8 through 1/2 inch; and 0.030 inch for 5/16 inch and smaller tubes. The bars shall have a hardness of 47 to 57 on the Rockwell C scale or be casehardened to a minimum depth of 0.003 inch with a surface hardness of 84 to 91 on the Rockwell 15-N scale. Micro hardness testing is acceptable and shall be equivalent to the Rockwell Hardness requirements. In order to cover the range of tubing specified, tool size J (see Table I) may be furnished with two pairs of die bars.

3.13.3 Compression assembly. The compression assembly shall provide the means for applying the necessary pressure for the flaring operation and shall consist essentially of a yoke, pressure screw and handle, and a flaring cone.

3.13.3.1 Type I, class 1, style B, yoke. The yoke shall have a threaded hole to receive the pressure screw, and the yoke construction shall permit ready alignment over the tube-gripping holes in the hinged bars, so that finished flares will meet the concentricity requirements specified in 4.4.1.1. Both hinged bars shall be supported the same height within 0.003 inch during the flaring operation.

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3.13.3.2 Pressure screw and handle. The pressure screw shall provide for the mounting of a swiveling flaring cone which does not rotate when the screw is turned during the flaring operation. A T-type sliding handle of adequate length to produce the required torque shall be provided on the opposite end. The handle shall not become detached from the pressure screw during normal usage. When specified (see 6.2), the screw shall have an extended additional head 1/4 inch square which permits use of a 1/4 inch ratchet wrench for working in close quarters. All pressure screws having the extended 1/4 inch square head shall have the T-type sliding handle also.

3.13.3.3 Flaring cone. The flaring cone shall be of the swiveling type and shall be mounted on the pressure screw in such manner that it will revolve freely, but will not become detached under hard usage. The conical face shall have a maximum surface roughness of 32 microinches. The cone shall have hardness of 51 to 62 on the Rockwell C scale or shall be casehardened to a depth of 0.010 to 0.020 inches and show a hardness of 86 to 91 on the Rockwell 15-N scale.

3.13.4 Type I, class 2, style A, illustration. Type I, class 2, style A, flaring tools shall conform to Table I for the tool letter specified and shall be similar to Figure 3 (see 6.2).

TABLE I. Type I, class 2, style A, flaring tool, screw-actuated, swivel cone, hinged dies

Tool letter	Range of tool, outside diameter of tubing in inches											
A	-	-	-	-	-	-	-	-	3/4	7/8	1	-
B	-	-	-	-	-	-	-	5/8	-	7/8	-	1-1/8
F	-	3/16	1/4	5/16	3/8	-	1/2	5/8	-	-	-	-
J*	1/8	3/16	1/4	5/16	3/8	7/16	1/2	5/8	3/4	-	-	-
J**	-	3/16	1/4	5/16	3/8	-	1/2	-	-	-	-	-
J***	-	3/16	1/4	5/16	3/8	-	1/2	5/8	3/4	-	-	-

* Single flaring.
** Double flaring.
*** Double flaring with swaging adapters.

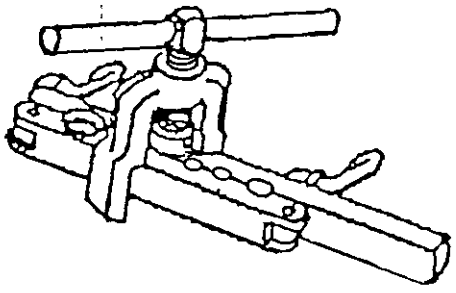


Figure 3. Type I, class 2, style A, flaring tool, screw-actuated, swivel cone, hinged dies.

3.14 Type I, class 3, flaring tool, screw-actuated combination, self-contained. Class 3 flaring tools shall be suitable for producing high-quality flares having a 74° included angle. The tools shall be capable of single flaring of all grades of aircraft tubing including one-quarter hard corrosion resisting steel, sizes 1/8 inch through 3/4 inch, in accordance with MS 33584. When specified (see 6.2), the tool shall produce 90° flares on soft copper and aluminum tubing in outside diameters 1/8 inch through 3/4 inch to fit standard S.A.E. fittings. The tool shall also be capable of double-lap flaring of mild aluminum tubing, sizes 3/16 inch through 3/4 inch. The double-lap flaring of tubing sizes 3/16 inch through 3/8 inch shall be in accordance with MS 33583.

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3.14.1 Type I, class 3, design. The tool shall be designed primarily for maintenance use and flares shall be made with a minimum application of torque capable of being exerted by an average mechanic working in a restricted position such as inside an aircraft wing. The tool shall be capable of withstanding the harder usage common in field service and shall operate adequately without permanent distortion or misalignment. The tool shall incorporate tube-setting stops or other positive means for flaring the tubing to the proper diameter. The first operation punches for double-lap flaring and the finish flaring cone shall be self-contained in the tool. The tool shall have a vise-grip attachment so that it can be used as a bench tool.

3.14.1.1 Material and hardness. The flaring cone and first operation punch shall be made from high grade carbon bar steel (S.A.E. 1020 minimum). The cone and punch shall be casehardened to a depth of at least 0.040 inch and have a hardness of at least 89 on the Rockwell 15-N scale.

3.14.2 Die (tube clamping) blocks. Die blocks shall be made of bar steel or powdered molded steel. When made of bar steel, die blocks shall have a minimum hardness of 47 on the Rockwell C scale or shall be casehardened to a minimum depth of 0.015 inch and show a minimum surface hardness of 84 on the Rockwell 15-N scale. When made of powdered molded steel, die blocks shall be carbonitrated and shall have a minimum hardness of 65 on the Rockwell 15-N scale.

3.14.3 Carrying case for type I, class 3. An adequate case (see 3.9) for the tool with compartments to accommodate all component parts shall be furnished. The case shall have a hinged lid with a positive catch.

3.14.4 Type I, class 3, weight. The weight of the tool including any one set of die blocks and the vise grip attachment shall not exceed 8 pounds. The weight of the tool complete with all components and the carrying case shall not exceed 10-1/2 pounds.

3.14.5 Type I, class 3, illustration. Type I, class 3, flaring tools shall be similar to Figure 4.

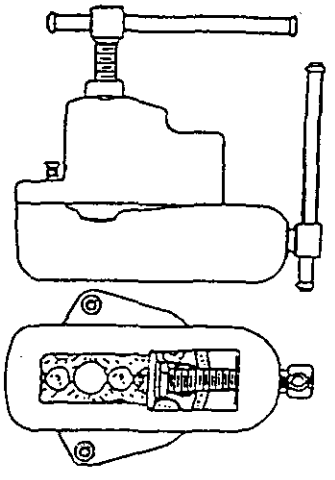


Figure 4. Type I, class 3, flaring tool, screw-actuated, combination self-contained.

3.15 Type II, class 1, flaring tool, impact-actuated, tube coupling. Class 1 flaring tools shall consist of a guide sleeve adapter and a flaring punch. The 74° tool shall flare tubes conforming to flare dimensions of MS 33584.

3.15.1 Guide sleeve adapter. The adapter shall be either threaded externally at one end for engaging an AN 818, or equal, triple-type tube coupling nut, or shall be recessed or counterbored for receiving an MS 20819, or equal, sleeve through a "side-access" opening. The side-access type of adapter shall have a spring clip or other suitable means of retaining the sleeve after it has been inserted in the adapter. Both types of adapters shall have a reamed hole to provide a free sliding fit for the flaring punch. The reamed hole shall be concentric with the locating threads or the locating recess within 0.005 inch total indicator reading.

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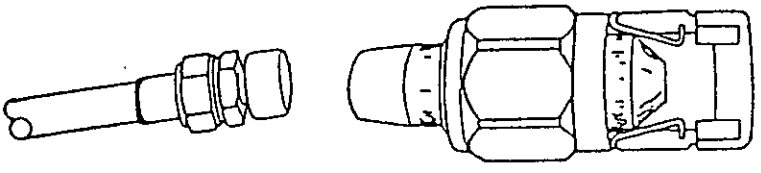
3.15.2 Type II flaring punch. The punch shall be cone-shaped at one end, with the striking head integral with or permanently fastened at the opposite end. The punch shall have a hardness of 37 to 52 on the Rockwell C scale. The striking head shall resist mushrooming or chipping and the cone shall not score during hard usage. The conical face shall have a maximum surface roughness of 32 microinches and the departure from straightness shall not exceed 0.002 inch per 1 inch length. The cone and the body shall be concentric within 0.005 inch total indicator reading.

3.15.3 Use data. Class 1 flaring tools shall utilize the triple-type sleeve in the coupling unit as the female die in the flaring operation, and shall be capable of flaring tubing which has a bend immediately adjacent to its end.

3.15.4 Type II, class 1, illustration. Type II, class 1, flaring tools shall conform to Table II for the tool number specified (see 6.2) and shall be similar to Figure 5 (A) or (B).

TABLE II. Type II, class 1, flaring tool, impact-actuated, tube coupling

Tool number	Applicable tube fitting	Included angle of flaring punch ± 1/2°	Tubing size nominal O.D.	Material and maximum wall thickness of tubing to be flared		
				Soft copper or aluminum	Steel fully annealed	Corrosion-resisting steel fully annealed
		Degrees	Inches	Inch	Inch	Inch
4A	AN	74	1/4	0.035	0.028	0.022
5A	AN	74	5/16	0.035	0.028	0.022
6A	AN	74	3/8	0.042	0.032	0.028
8A	AN	74	1/2	0.049	0.035	0.032
10A	AN	74	5/8	0.049	0.035	0.032
12A	AN	74	3/4	0.065	0.049	0.035
16A	AN	74	1	0.065	0.049	0.035
20A	AN	74	1-1/4	0.065	0.049	-
24A	AN	74	1-1/2	0.065	0.049	-



(a) (b)

Figure 5. Type II, class 1, flaring tool, impact-actuated, tube coupling.

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3.16 Type II, class 4 flaring tool, impact-actuated, hinged blocks. Class 4 flaring tools shall consist of die bars and a yoke assembly. The tool shall flare tubes conforming to flare dimensions of MS 33584.

3.16.1 Type II, class 4 die bars. The die bars shall have seven countersunk tube-clamping holes and shall be hinged at one end in such a manner as to insure positive alignment of each size hole and easy removal of the tubing after flaring. Provisions for centralizing each tubing hole with the flaring punch in the yoke assembly shall be incorporated in one or both of the die bars. The countersinks shall be concentric with the holes within 0.005 inch total indicator reading. The amount of potential squeeze between die bars shall be sufficient to clamp tubing securely, but shall be held to a minimum to prevent excessive tube deformation, and in any case shall not exceed the following limits: 0.040 inch for 5/8 inch and larger; 0.035 inch for 3/8 through 1/2 inch; and 0.020 inch for 5/16 inch and smaller tubes. The die bars shall be casehardened to a minimum depth of 0.015 inch and show a surface hardness of 78 to 86 on the Rockwell 15-N scale. Micro hardness testing is acceptable and shall be equivalent to the Rockwell Hardness requirements.

3.16.2 Yoke assembly. The yoke assembly shall consist essentially of a yoke, flaring punch, and a clamping screw.

3.16.2.1 Type II, class 4 yoke. The yoke shall have a reamed hole perpendicular to the countersunk side of the die bars and shall provide a sliding fit for the flaring punch. The side of the yoke shall have a threaded hole to receive the clamping screw. Provisions shall be made for a spring catch or other means to hold the flaring punch in a raised position while sliding the yoke from one location to another. The yoke shall have a feature for self-centering over each hole when the clamping screw has been tightened ready for flaring.

3.16.2.2 Class 4 flaring punch. The punch shall conform to 3.15.2.

3.16.2.3 Clamping screw. The clamping screw shall be provided with a T-type sliding handle of adequate length to produce the proper torque when hand tightened. The sliding handle shall not become detached from the screw under normal usage. The point of the screw shall be oval-shaped and hardened to a hardness of 47 to 52 on the Rockwell C scale, or to 78 to 86 on the Rockwell 15-N scale when casehardened to prevent scoring when tightened against the die bars.

3.16.3 Type II, class 4, illustration. Type II, class 4, flaring tools shall conform to Table III and shall be similar to Figure 6.

TABLE III. Type II, class 4, flaring tool, impact-actuated, hinged blocks.

Applicable tube fitting	Included angle of flaring punch ± 1/2°	Tubing size nominal O.D.	Material and maximum wall thickness of tubing to be flared		
			Soft copper or aluminum	Steel fully annealed	Corrosion-resisting steel fully annealed
	Degrees	Inch	Inch	Inch	Inch
AN	74	1/8	0.035	0.035	0.028
AN	74	3/16	0.035	0.035	0.028
AN	74	1/4	0.065	0.049	0.049
AN	74	5/16	0.065	0.049	0.049
AN	74	3/8	0.065	0.065	0.049
AN	74	1/2	0.083	0.065	0.065
AN	74	5/8	0.095	0.072	0.065

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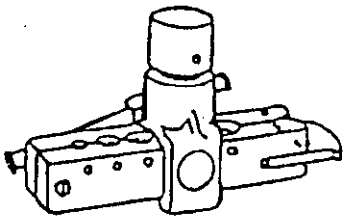


Figure 6. Type II, class 4, flaring tool, impact-actuated, hinged blocks.

3.17 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, adherent, continuous, and not stained or discolorerd. Fasteners shall be firmly secured and show no evidence of deformatin, cross threading, or hazardous burrs. Adhesives and lubricants, adequate for their intended purpose, shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners or sufaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

3.18 Metric product. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Federal Specification are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, the contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for ensuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Visual examination. The end items shall be examined for the defects listed in Table IV. The lot size shall be expressed in terms of flaring tools. The sample unit shall be one flaring tool. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 4.0 for total (major and minor combined) defects.

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TABLE IV. Classification of visual defects

Examine	Defect	Major	Minor
Construction and Workmanship	Type, class, style, size, tool number, diameter of pilot, and angle of flare not as specified	101	
	Tool incomplete	102	
	True diameter of tube-gripping holes not within the specified tolerance (type I, class 1, styles A and B)	103	
	True diameter of tube-gripping holes not within the specified tolerances or holes not countersunk (type II, class 4)	104	
	Tube-gripping holes not countersunk, or not provided with a means of preventing the tubes from slipping (type I, class 2, style A)	105	
	Pressure screw defective, sliding handle not prevented from becoming detached, handle not of sufficient length, or extended square head (if required) missing or not as specified (type I, class 2, style A; type I, class 3)	106	
	Driving lugs on underside of shaft head and on the pressure sleeve not securely fastened, or lugs fail to drive against each other (type I, class 1, styles A and B only)	107	
	Failure to provide a means for allowing the flaring head to rotate in the same plane (for truing the flare) (type I, class 1, styles A and B)	108	
	Flaring cone defective, not free to revolve, or not properly mounted on end of pressure screw (type I, class 2, style A)	110	
	Flaring cone defective, not securely attached to pressure screw, not finished as required, or cone not aligned with die blocks (type I, class 3)	111	
	Not provided with a means of aligning the flaring punch over each tube-gripping hole (type II, class 4)	112	
	Not provided with tube setting stops or other means for flaring the tubing to the proper diameter (type I, class 3)	113	
Materials	Defective; evidence of pits, nodules, burrs, or cracks	114	
Components and Assemblies	Any component not aligned with mating part	115	
Clamping device	Not securely held in place or fails to swivel easily (type I, class 1, styles A and B only)	116	
	Not securely held in place, clamping device not easily swiveled, sliding handle not of sufficient length, or sliding handle not prevented from becoming detached (type I, class 3)	117	
	Clamping screw head or shaft head defective, not knurled, or not as specified (type I, class 1, styles A and B)	118	

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TABLE IV. Classification of visual defects (cont'd)

Examine	Defect	Major	Minor
Components and Assemblies Clamping device (cont'd)	Clamping screw defective, point not rounded, sliding handle not of sufficient length, or sliding handle not prevented from becoming detatched (type II, class 4 only)	119	
Die blocks/bars	Thickness of die blocks exceeds the specified maximum (type I, class 1, styles A and B)	120	
	Die bars not supported the same height (by the yoke) within the maximum allowable (type I, class 2, style A only)	121	
	Wing nuts fail to clamp bars securely in place, or fail to permit easy removal of the tubing (type I, class 2, style A only)	122	
	Triple type sleeve (die) defective or not aligned (type II, class 1)	123	
Driving lugs	Driving lugs on underside of shaft head and on the pressure sleeve not securely fastened, or lugs fail to drive against each other (type I, class 1, styles A and B only)	124	
Yoke assembly	Fails to permit ready alignment over the tube-gripping holes (type I, class 1, styles A and B; type I, class 2, style A)	125	
Deburring reamer (if required)	Missing, defective, yoke not easily positioned, reamer defective, handle not secure, or tool not as specified (type I, class 1, styles A and B)	126	
Vise grip attachment	Not provided (type I, class 3)	127	
Gage	Missing or defective (type I, class 1, style A only)	128	
Punch	Defective, chipped, or striking head not permanently fastened (type II, class 1; type II, class 4)	129	
	Flaring punch not provided with a spring catch to hold it in the raised position (type II, class 4)	130	
Adapter	Threads defective, torn, stripped, crossed, or not as specified (type II, class 1)	131	
	Hole fails to provide a free sliding fit for the punch (type II, class 1)	132	
Carrying Case	Defective, fails to provide compartments for all components, hardware (handle, hinges, and catch) defective or inoperable, or painting defective or incomplete (type I, class 3)		201
Marking	Manufacturer's name or trademark missing, illegible, incorrect, or not permanently marked		202

4.3.2 Dimensional and weight examination. The end items shall be examined for nonconformance to dimensional and weight requirements specified herein. The lot size shall be expressed in terms of flaring tools. The sample unit shall be one flaring tool. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.4 Testing. The end items shall be tested for the characteristics specified in 4.4.1, 4.4.2, and 4.4.3. Air Force purchases shall also be tested for the characteristics specified in 4.4.4.1. The lot size shall be expressed in units of flaring tools. The sample unit shall be one flaring tool. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.1 Performance test. Each of the tools selected in accordance with 4.2 shall flare one end of at least two pieces of each size of tubing specified. Tubing of the material listed in Table V, and having a wall thickness equal to or greater than that shown in Table V opposite the applicable type, class, and style of tool shall be used for this test. When a deburring reamer is furnished (see 3.10.5), it shall be used to deburr the tubing used for testing the parent tool. The tubing end to be flared shall be cut off clean, square, and the sharp edges smoothly deburred. All flares shall be smooth and free from cracks and pit marks. During and after the performance test, there shall be no evidence of scoring or scratching of the cone or rollers which would affect the use of the flare; metal adhering to the cone or rollers; failure of any component to function properly; difficulty in removal or insertion of replaceable parts; or permanent deformation or damage of components. In addition, the inspection of all tubes flared by type I, class 1, styles A and B; type II, classes 1 and 4 tools shall show no cutting, scratching, or marring of the tubing surfaces which may under vibration and strain cause the tubing to fail in service.

TABLE V. Material and nominal dimensions of tubing used in the performance test for flaring

Operation performed	Tool designation			Outside diameters of tubing in inches													Material
	Type	Class	Style	1/8	3/16	1/4	5/16	3/8	7/16	1/2	5/8	3/4	7/8	1	1-1/8	1-1/4	1-1/2	
				Wall thickness (inches)													
Single flare	I	1	A	-	0.032	0.032	0.032	0.032	-	0.035	0.035	-	-	-	-	-	-	Cold drawn corrosion-resisting steel
Single flare	I	1	B	-	-	-	-	-	-	-	-	0.065	0.065	0.065	-	0.083	-	Annealed corrosion-resisting steel
Single flare	I	2	A	0.020	0.028	0.035	0.035	0.035	0.035	0.035	0.049	0.049	0.049	0.058	0.058	-	-	Steel annealed
Double-lap flare	I	2	A	-	0.028	0.028	0.032	0.032	-	0.032	0.032	0.032	-	-	-	-	-	Steel annealed
Single flare	I	3	-	-	0.028	0.028	0.032	0.032	-	0.035	0.035	-	-	-	-	-	-	Cold drawn corrosion-resisting steel
Double-lap flare	I	3	-	-	-	0.028	0.028	0.032	-	-	-	-	-	-	-	-	-	Mild aluminum
Single flare	II	1	-	-	-	0.028	0.028	0.032	-	0.035	0.035	0.049	-	0.049	-	0.049	0.049	Steel annealed
Single flare	II	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	Steel annealed

4.4.1.1 Type I, class 1, styles A and B; class 3; type II, classes 1 and 4. The 74° single-lap flares produced by these tools shall conform to Figure 7, as specified. Concentricity shall also be in accordance with the applicable figure.

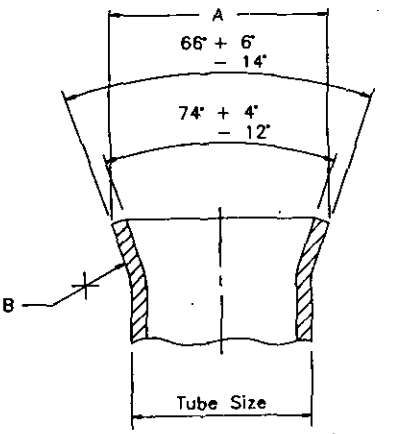


Figure 7. 74° single-lap flare.

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Tube size nominal O.D.	"A" diameter		"B" radius ± .030
3/16	0.302	+ 0.000 - 0.033	0.125
1/4	0.359	+ 0.007 - 0.022	0.125
5/16	0.421		0.140
3/8	0.484		0.187
1/2	0.656		0.187
5/8	0.781		0.312

Flare shall be square with centerline of tube within 3°.
Flare shall be concentric with O.D. of tube within 0.037 TIR.
Flare shall be free of cracks and pit marks.
For use with standard flared tube fittings.
Dimensions are in inches.

Figure 7. 74° single-lap flare (cont'd).

4.4.1.1.1 Type I, class 3. The 74° double-lap flares produced by these tools shall conform to MS 33583.

4.4.1.2 Type I, class 2, style A. The 90° single flares produced by these tools shall be in accordance with Figure 8, except they shall be concentric with the outside diameter of the tubing within 0.012 inch total indicator reading, and the included angle inside the flare shall be between 88° and 92°.

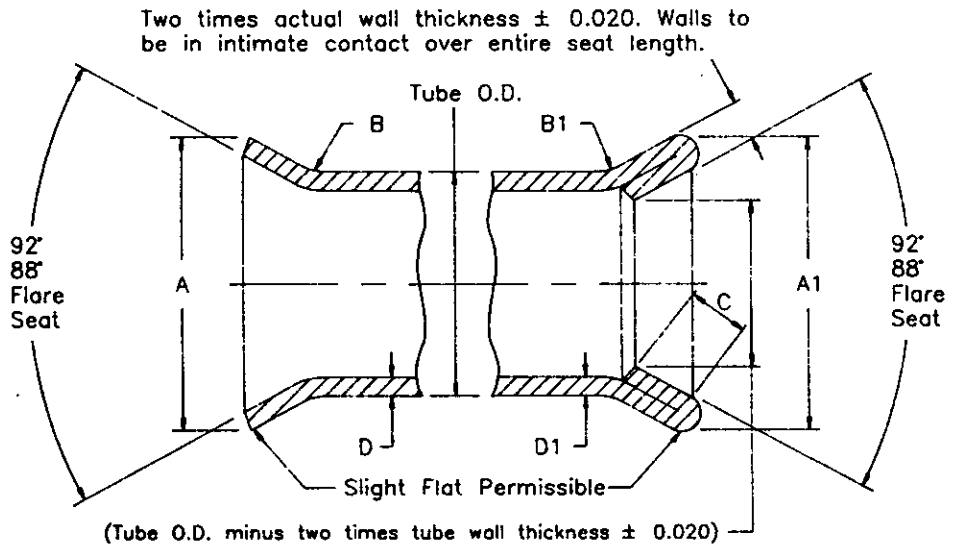


Figure 8. Flares produced by type I, class 2, style A flaring tools.

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Nominal Tube O.D.	"A" Single flare diameter		"A1" Double flare diameter		"B" Single flare radius	"B1" Double flare radius	"C" Double flare coined seat length
Inch	Inch		Inch		Inch	Inch	Inch
	Max.	Min.	Max.	Min.	+ 0.150 - 0.010	+ 0.150 - 0.010	Min.
1/8	0.181	0.171	0.213	0.198	0.040	-	-
3/16	0.249	0.239	0.280	0.265	0.040	0.040	0.040
1/4	0.325	0.315	0.360	0.345	0.040	0.040	0.040
5/16	0.404	0.388	0.425	0.410	0.040	0.040	0.062
3/8	0.487	0.471	0.500	0.485	0.040	0.040	0.062
7/16	0.561	0.545	0.570	0.555	0.040	-	-
1/2	0.623	0.607	0.640	0.625	0.040	0.040	0.062
5/8	0.748	0.732	0.772	0.757	0.040	-	-
3/4	0.916	0.900	0.912	0.897	0.040	-	-

Single flares shall be concentric with O.D. of tubing within 0.020 TIR.
Double flares shall be concentric with O.D. of tubing within 0.030 TIR.

Figure 8. Flares produced by type I, class 2, style A, flaring tools (cont'd).

4.4.1.3 Type I, class 2, style A. The 90° double-lap flares produced by these tools shall meet the requirements specified for 90° single-lap flares (see 4.4.1.2).

4.4.1.4 Type II, class 1. The 74° flares produced by these tools shall be concentric with the outside diameter of the tubing within 0.008 inch total indicator reading.

4.4.2 Hardness test. Each sample tool selected in accordance with 4.2 shall be tested in accordance with ASTM E 18.

4.4.3 Surface roughness. Surface roughness specified in 3.11.2 shall be tested in accordance with ANSI B 46.1.

4.4.4 Tests. When specified for Air Force purchases, a hydraulic pressure test and a gas pressure test shall be conducted in addition to the tests specified in 4.4.1 through 4.4.3 for the types, classes, and styles indicated below.

4.4.4.1 Hydraulic pressure test. Each sample 1/2 inch size of type I, class 1, style A; and type I, class 3 flaring tool selected in accordance with 4.2 shall be tested as follows: Flare 1/2 inch O.D. by 0.049 inch wall thickness annealed corrosion-resisting steel hydraulic tubing on both ends. Assemble one sample of tubing for each tool selected with coupling nuts (Drawing AN 818) and sleeves (MS 20819) to a union (Drawing AN 815) with a cap assembly (Drawing AN 929) attached. Connect tubing assembly together using a wrench torque of 450 inch-pounds. The tubing assembly shall then be connected to a hydraulic pressure source (oil or clear water) and subjected to a pressure of 6,000 pounds per square inch. There shall be no evidence of any leakage of the flared tube connections.

4.4.5 Possible test failures. Possible test failures are as shown in Table VI.

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TABLE VI. Possible test failures

Characteristic	Possible failure
Hardness:	Hardness of die blocks, flaring head, or rollers, not within the specified values on the Rockwell C or 15-N scale (type I, class 1, styles A and B; type I, class 3)
	Hardness of cone or die bars (dies or die blocks) not within the specified value on the Rockwell C or 15-N scale (type I, class 2, style A)
	Hardness of punch die bars or die blocks not within the specified values on the Rockwell C or 15-N scales (type II, classes 1 and 4)
Finish:	Surface roughness of tube-gripping holes, flaring head rollers, cone, conical face of the punch
Straightness of rollers:	Rollers depart from straightness by more than the maximum allowable (type I, class 1, styles A and B)
Straightness:	Cone face departure from straightness exceeds the maximum allowable (type II, classes 1 and 4)
Concentricity:	Concentricity of reamed holes, tube-clamping holes, and countersinks, or concentricity of cone face with the body of the punch not within the specified total indicator readings (type II, classes 1 and 4)
Weight:	Weight of complete tool with carrying case exceeds the maximum allowable (type I, class 3)
Bar separation:	Separation between die bars exceeds the specified maximum when standard drill rod is clamped between them (type II, class 4)
Block squeeze:	Separation between mating die blocks not within the specified maximum and minimum values when the dimensions are measured as indicated on Figure 9
Performance testing:	Failure of tool to flare the specified tubes, or excessive torque required to flare tubes, or evidence of metal adhering to rollers after the test. Flare produced by tool fails to conform with the specified requirements, evidence of cracks or scratches in the tubing, or outside diameter of flare not concentric with outside diameter of tube within the specified total indicator reading. Deburring tool (if required) fails to perform its required function, die blocks distort the tubing more than the maximum allowable (type I, class 1, styles A and B; type I, class 2, style A)
	Failure of tool to flare the specified size tube, striking head mushrooms or chips, or cone is scored during the test. Flare produced by the tool fails to conform with specified requirements, evidence of cracks or pits in the tubing, or outside diameter of flare not concentric with outside diameter of tube within the specified total indicator reading (type II, classes 1 and 4)

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A Tube Size In.	B + B1. In. Actual	C Maximum In. A -(B + B1)
3/4	-	0.038
7/8	-	0.038
1	-	0.038
1-1/4	-	0.038

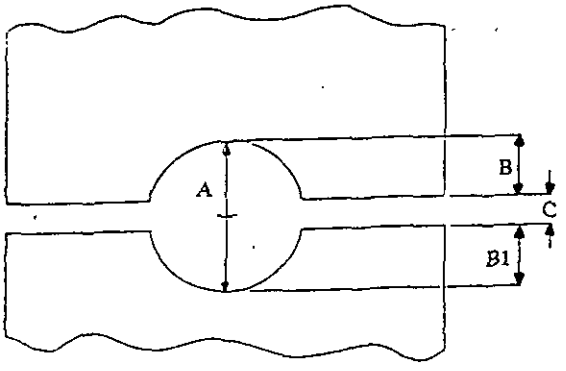


Figure 9. Measurement of separation between die blocks.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.2).

6. NOTES

6.1 Intended use.

6.1.1 Type I, class 1, style A and type I, class 3. Type I, class 1, style A and type I, class 3 flaring tools are used for producing high quality flares on corrosion-resisting steel tubing in both cold drawn and annealed state for aircraft hydraulic fuel, oil, and air lines. Type I, class 3 flaring tools are also used for producing high quality double-lap flares on mild aluminum tubing used in aircraft.

6.1.2 Type I, class 1, style B. Type I, class 1, style B flaring tools are used for producing high quality flares on steel and annealed corrosion-resisting steel tubing for hydraulic and other equipment requiring strong flares on various weights of tubing.

6.1.3 Type I, class 2, style A. Type I, class 2, style A flaring tools are used for producing single- and double-lap flares of a general purpose quality on soft copper and steel tubing for automotive fuel, oil, and water lines, and other general purpose equipment.

6.1.4 Type II, classes 1 and 4. Type II, classes 1 and 4 flaring tools are used for producing good quality flares on copper, aluminum, and soft steel tubing for hydraulic, steam, refrigeration, and instrumentation lines.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style if applicable, of tool required (see 1.2).
c. Finish, if a choice is required (see 3.3).
d. Tool number or letter, as applicable, of tool required (see Tables I and II).
e. Precision feed control deburring reamer, if required for type I, class 1, style A flaring tool (see 3.11.1 and 3.11.5).
f. Boxes, if required for type I, class 2, style A flaring tools (see 3.13).
g. Adapter set for double-lap flares, if required for type I, class 2, style A flaring tool (see 3.13.1).
h. Extended head 1/4 inch square, if required for type I, class 2, style A (see 3.13.3.2).
i. Flares having 90° included angle, if required for type I, class 3 flaring tool (see 3.14).
j. Level of preservation, packaging, and packing required (see 5.1).
k. Marking, if other than specified.

6.2.1 When ordering repair parts, specify the following:

a. The manufacturer's name, or trademark of the tool for which the parts are required.
b. Type, class, and style of tool, if applicable.
c. Tool letter or number, if applicable (see Tables I and II).
d. Included angle of flare in degrees and wall thickness of tubing, if applicable.
e. The name or description of the part, and size if applicable.
f. Quantity of each part required.

6.3 Supersession data. The classification of flaring tools in this specification, corresponding to those covered in Federal Specification GGG-F-451A are as follows:

GGG-F-451A	GGG-F-451D
Type I. Flaring, screw-actuated	Type I
Class 1 - Roller Cone	Class 1
Style A - Attached flare gage	Style A
Style B - Without attached gage	Style B
Class 2 - Swivel cone	Class 2
Style A - Hinged dies	Style A
Style B - Block dies	Deleted
Style C - Dowelled dies	Deleted
Style D - Indexing dies	Deleted
Class 3 - Combination, self-contained	Class 3
Type II. Flaring, impact-actuated	Type II
Class 1 - Tube coupling	Class 1
Class 2 - Hand block	Deleted
Style A - Single flare	Deleted
Style B - Double flare	Deleted
Class 3 - Vise blocks	Deleted
Class 4 - Hinged blocks	Class 4
Class 5 - Removable dies	Deleted
Type III. Reading, rotary	Deleted
Class 1 - Bench	Deleted
Class 2 - Portable	Deleted

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MILITARY INTERESTS:

Military Custodians:

Army - GL
Air Force - 99

Review Activities:

Army - AV
Air Force - 84

User Activities:

Army - AT, ME, CE
Navy - YD, MC

Military Coordinating Activity:

Army - GL

Preparing Activity:

GSA - FSS

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